

PRACTICE TEST 2 ANSWER KEY

SECTION I: Multiple Choice

1. b
2. c
3. c
4. b
5. b
6. c
7. b
8. c
9. b
10. c
11. b
12. b
13. c
14. b
15. b
16. c
17. c
18. b
19. b
20. b

SECTION II: True/False

21. False
22. True
23. False
24. True
25. False
26. True
27. True
28. True

29. False

30. False

SECTION III: Short Answer

31. Pathway of light through eye (10 points):

- Cornea (2 pts)
- Anterior chamber/aqueous humor (2 pts)
- Pupil/lens (2 pts)
- Vitreous humor (2 pts)
- Retina (2 pts)

32. Hormone mechanisms (10 points): Lipid-soluble (5 pts):

- Pass through membrane
- Bind internal receptors
- Enter nucleus
- Affect gene transcription
- Slow but long-lasting effects

Water-soluble (5 pts):

- Bind membrane receptors
- Activate second messengers
- Trigger enzyme cascades
- Rapid response
- Short-term effects

33. Menstrual cycle hormones (10 points):

- FSH rises early in follicular phase (2 pts)
- Estrogen rises as follicle develops (2 pts)
- LH surge triggers ovulation (2 pts)
- Progesterone rises in luteal phase (2 pts)
- All hormones drop if no pregnancy (2 pts)

34. Sound transmission (10 points):

- Sound waves enter external ear (2 pts)
- Tympanic membrane vibrates (2 pts)
- Ossicles amplify vibrations (2 pts)
- Oval window receives vibrations (2 pts)

- Fluid movement in cochlea stimulates hair cells (2 pts)